CerebroTwin

A Personalized Digital Twin for Forecasting Orthostatic Cerebral Blood-Flow-Velocity Instability in Type 2 Diabetes

Hackathon Submission — October 2026

Team Members

Harish Pannuru — CSE — Lead ML & Full-Stack Architect

Pavan Sai Krishna — MECH — Physiological Systems & Dynamics Lead

Girish Chandra Kovuru — MECH — Biomedical Signal & Feature Engineer

Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram

Chennai, Tamil Nadu, India

Clinical Problem Statement

Orthostatic Cerebral Hypoperfusion in Type 2 Diabetes

- T2D patients develop autonomic neuropathy → impaired cerebral autoregulation
- Sit-to-stand posture change causes acute systemic blood pressure drop
- Healthy baroreflex compensates within seconds; T2D response is blunted & delayed
- Result: transient cerebral hypoperfusion → dizziness, syncope, falls, injury
- Brachial BP and HR alone cannot reflect cerebral hemodynamics
- TCD (Transcranial Doppler) measures MCAv directly — but is impractical for ambulatory use
- Unmet need: predict cerebral flow trajectory from non-cerebral bedside signals

Key Insight: Systemic cardiovascular signals encode latent information
about cerebral autoregulatory capacity that ML can decode

Healthcare Use Case & Research Scope

Non-Diagnostic Clinical Decision Support Framing

- Target Population: Adults with Type 2 Diabetes (T2D)
- Clinical Setting: Pre-operative, geriatric, endocrine clinics
- Use Case: Forecast patient-specific MCAv trajectory
- Risk Stratification: Flag delayed cerebral recovery

Research Scope Disclaimer

CerebroTwin is a proof-of-concept research tool — NOT a diagnostic device.
Forecasts are flagged as deviations from control distributions, not disease labels.

- Outputs are decision-support signals, not prescriptive treatment recommendations
- Counterfactual simulations illustrate physiological sensitivities, not clinical advice
- All validation uses strictly held-out subjects (GroupKFold) — no data leakage

Digital Twin Methodology

PhysioNet Dataset Alignment & Physiological Feature Extraction

- Data Source: PhysioNet 'Cerebral Vasoregulation in Elderly with Stroke' dataset
- Subjects: Healthy controls + stroke/T2D cohorts with concurrent multi-signal recordings
- Signals: Beat-to-beat arterial BP, ECG/HR, respiration, EtCO■, and TCD MCAv

econd

Non-Cerebral Features (Model Input)

Static: Age, Sex, T2D, HbA1c, HTN, Meds, Resting HR/MAP
HRV: SDNN, RMSSD, LF/HF ratio
Dynamic: ΔMAP/Δt, ΔHR/Δt slopes (0–5s post-stand)
Respiratory: Rate shifts, EtCO■ baseline & drop

Ground Truth (Validation Only)

TCD Middle Cerebral Artery velocity (MCAv)
Strictly withheld from model training inputs
Used only as supervised learning target
Subject-level isolation enforced

Machine Learning Architecture

Strict Subject-Level GroupKFold Cross-Validation (K=5)

Model 0: Mean Baseline

Population mean control trajectory

Reference benchmark for comparison

Model 1: Ridge Regression

L2-regularized linear model

Tests linear signal–cerebral relationship

Model 2: Quantile XGBoost

Multi-output gradient boosted trees

Quantile regression (10th/50th/90th)

90% prediction uncertainty intervals

GroupKFold (K=5): Every subject appears in TRAIN or TEST, never both.

Eliminates temporal & identity leakage — critical for medical ML validity.

Results & Validation Metrics

Quantile XGBoost on Held-Out Subject Folds

0.735

R^2

Coefficient of Determination

4.816%

MAE

Mean Absolute Error (pct MCAv drop)

4.475%

RMSE

Trajectory RMSE (0–60s window)

- All metrics computed on held-out test subjects (GroupKFold split)
- Model never sees TCD during training — pure non-cerebral → cerebral prediction
- Quantile heads provide 90% prediction intervals for clinical uncertainty quantification
- Recovery trajectory classified as Normal or Delayed based on 30-second threshold

SHAP Explainability

Autonomic & Vascular Feature Drivers

- SHAP (SHapley Additive exPlanations) via TreeExplainer for XGBoost
- Per-patient feature attribution — which inputs drive THIS patient's prediction?
- Top autonomic drivers: ΔMAP/Δt (BP drop slope), ΔHR/Δt (compensatory HR), LF/HF ratio
- Top vascular drivers: Resting MAP, baseline HRV (SDNN, RMSSD), EtCO■ reactivity
- Clinical modifiers: HbA1c level, T2D duration, β-blocker status

Clinical Interpretability

Every prediction is accompanied by a SHAP waterfall plot showing exactly which physiological features pushed the forecast toward risk or safety. This transparency is essential for clinician trust in AI-assisted decision support.

Counterfactual What-If Simulation

Clinician Decision Support Engine

- Interactive "What-If" engine in the Streamlit Clinician Dashboard
- Modify any clinical input and instantly see updated MCAv trajectory forecast
- Example: "What if HbA1c drops from 9.2% to 7.5% with medication?"
- Example: "What if a β-blocker is added to the current regimen?"
- Side-by-side trajectory comparison: Baseline vs. Counterfactual scenario
- Helps clinicians explore physiological sensitivities before real interventions

Simulator Capabilities

Adjust: Age, HbA1c, HTN status, Medications, Resting HR, MAP
Visualize: Predicted trajectory shift, nadir change, recovery time delta
Compare: Original vs. counterfactual with uncertainty bands overlay

CerebroTwin — Bridging Systemic Signals to Cerebral Insight

Thank you for reviewing our submission